PROS

- Sensores que miden
- Control que corrige
- Robots que ganan

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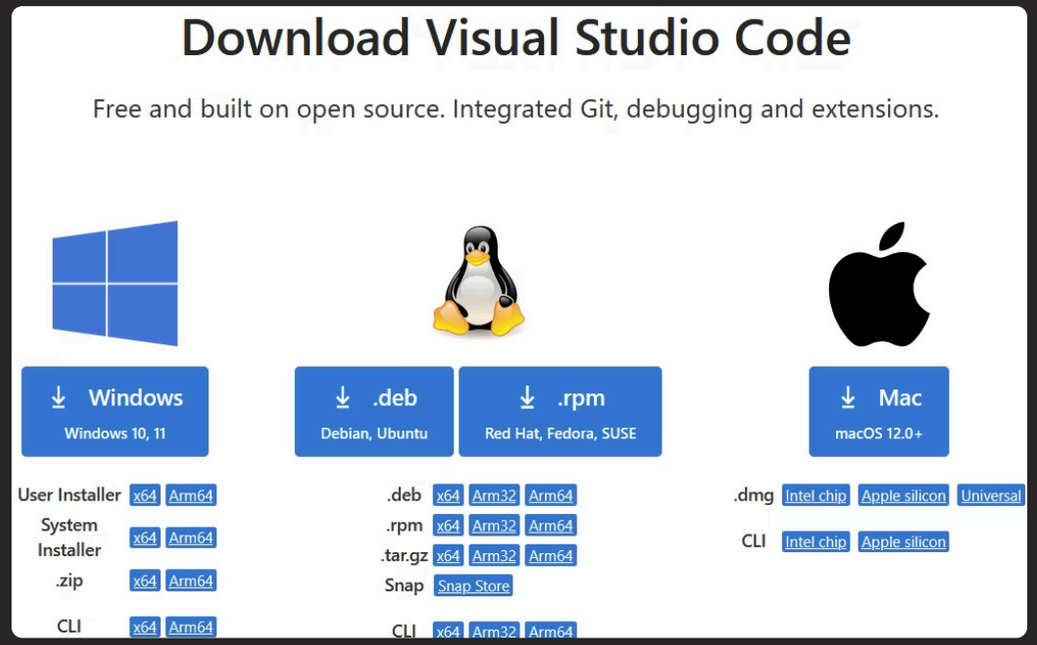
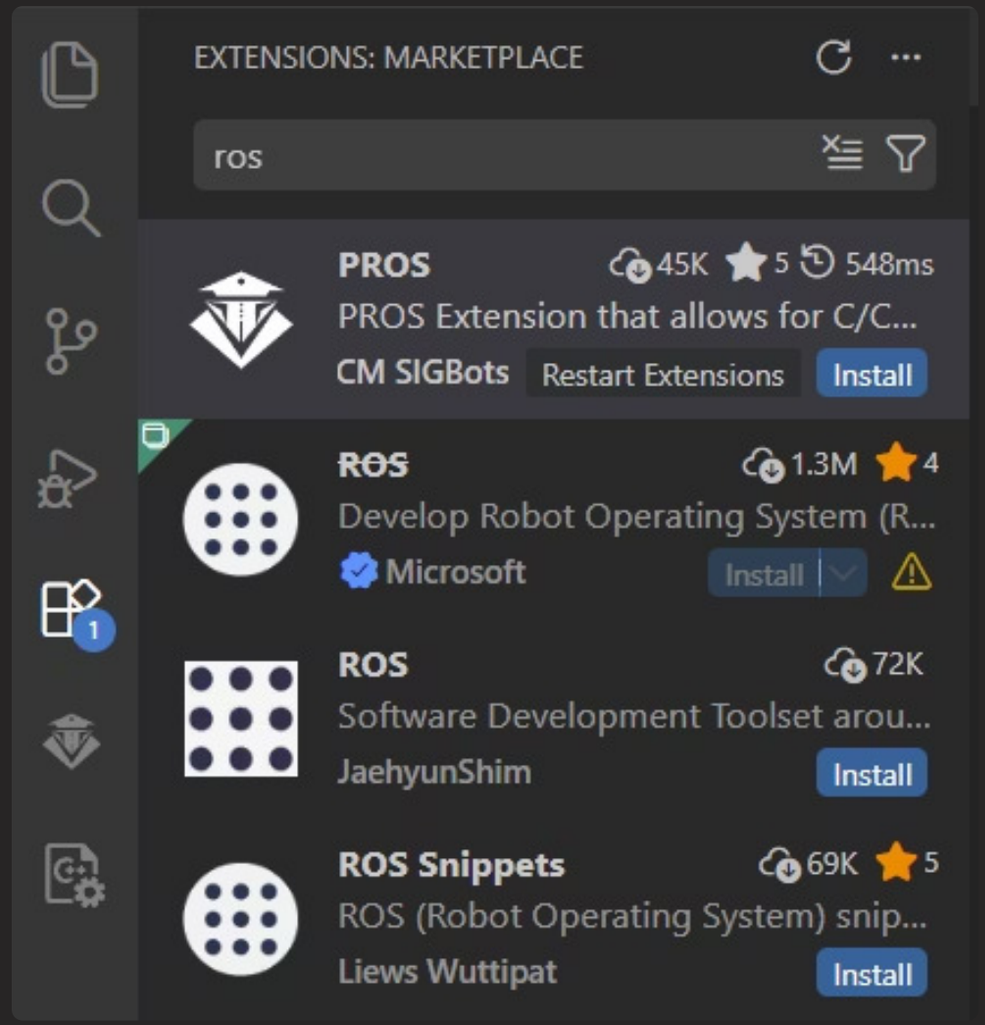
INSTALACIONES

Descargar VS Code



Abre extensiones y descarga PROS



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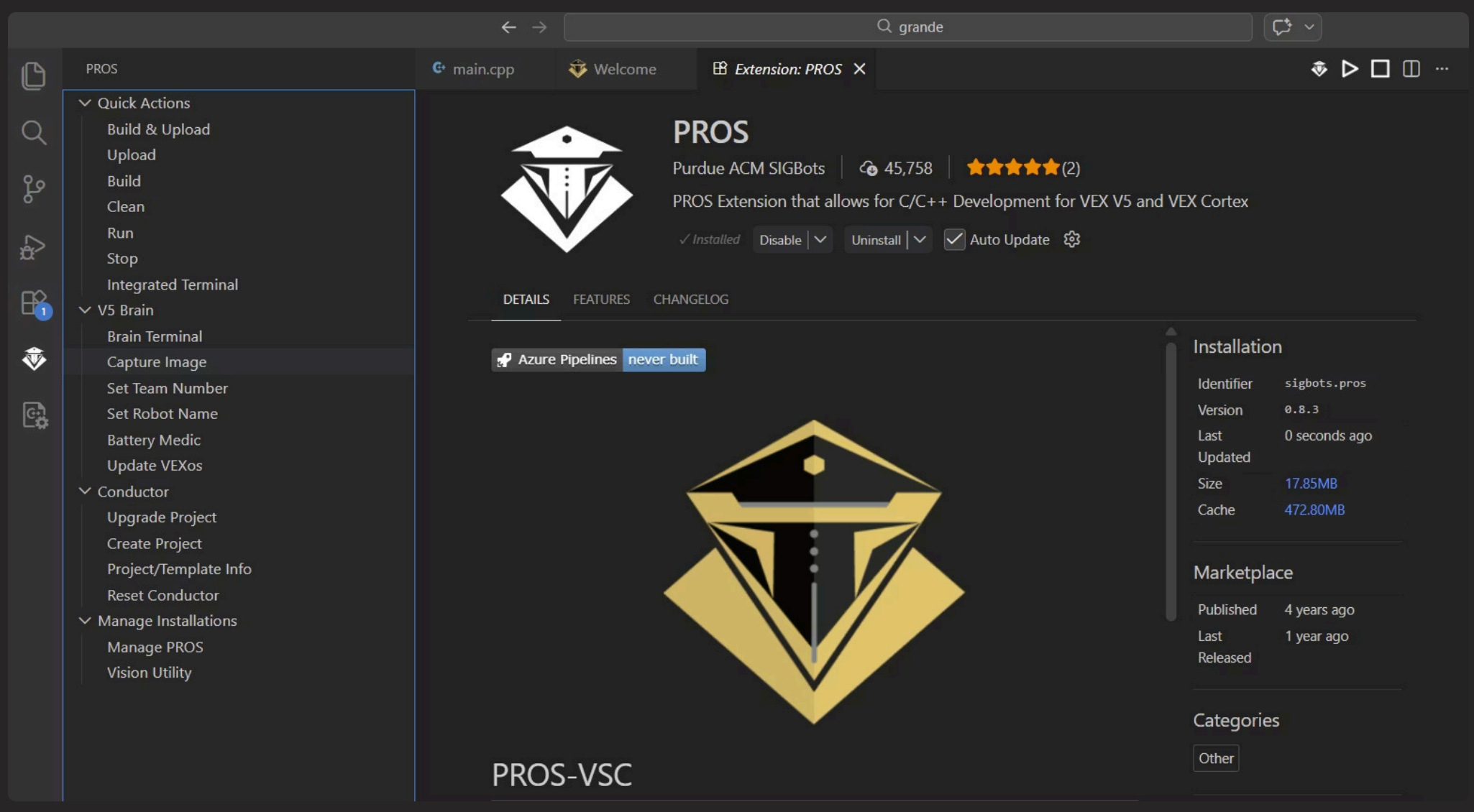
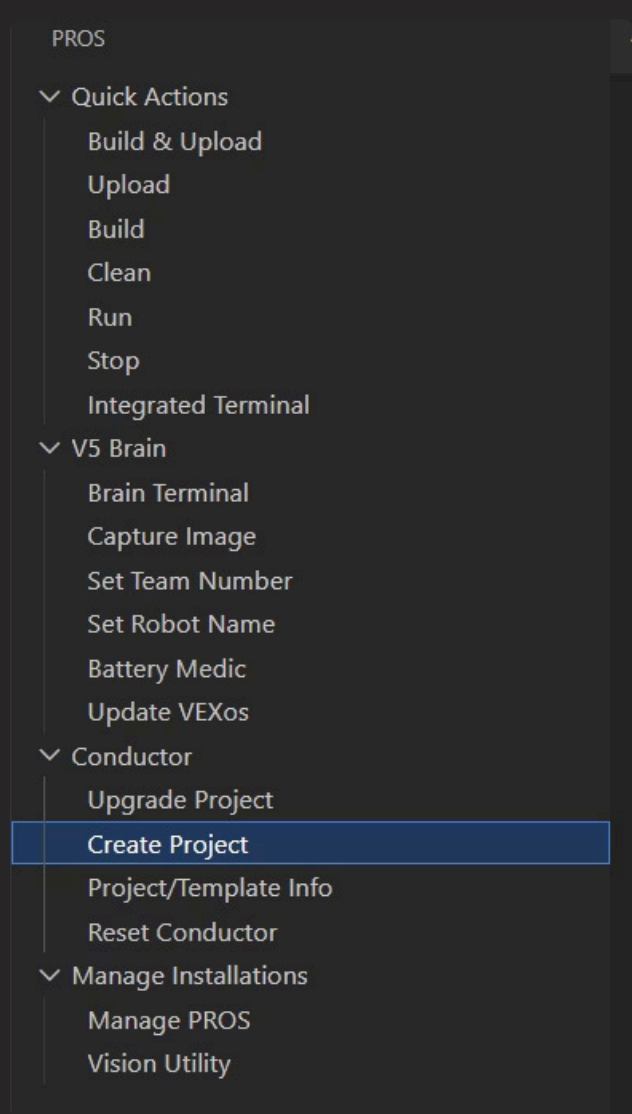
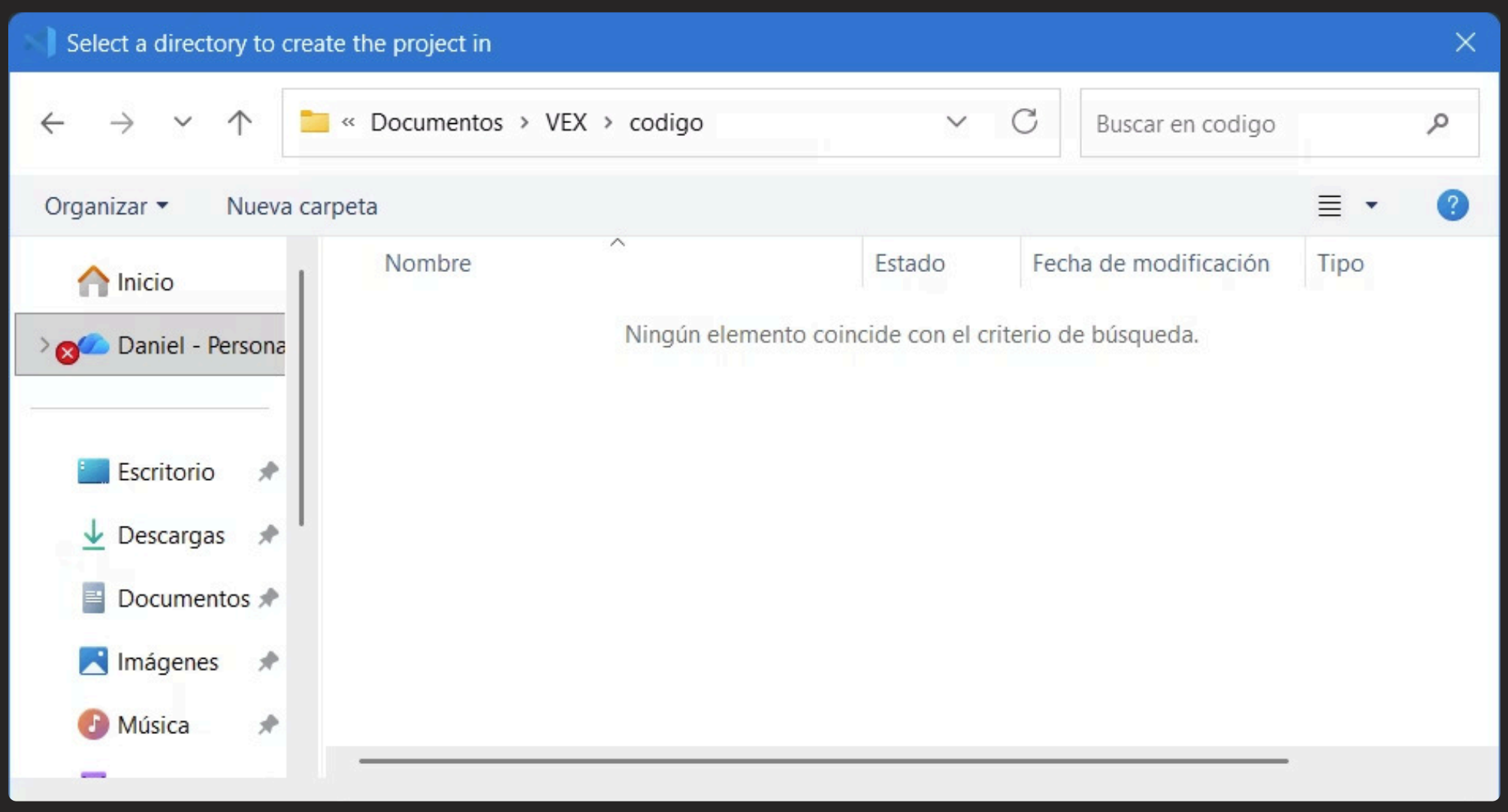




CREATE PROJECT



CREAR CARPETA

DOCS»VEX/VORTEX»CODE

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test_robot

main.cpp

Enter a name for the project (Press 'Enter' to confirm or 'Escape' to cancel)

PROS

Purdue ACM SIGBots | 45,758 | (2)

PROS Extension that allows for C/C++ Development for VEX V5 and VEX Cortex

Select the target device

v5

- v5
- cortex

Select Project Version

latest

- latest Recommended
- 4.2.2
- 4.2.1
- 4.1.2
- 4.1.1
- 4.1.0
- 3.8.3

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EXPLORER

TEST_ROBOT

- .d
- .vscode
- bin
- firmware
- include
- src
  - main.cpp
- .gitignore
- common.mk
- compile_commands.json
- Makefile
- project.pros

Welcome | main.cpp

src > main.cpp > on_center_button()

```cpp
#include "main.h"

/**
 * A callback function for LLEMU's center button.
 *
 * When this callback is fired, it will toggle line 2 of the LCD text between
 * "I was pressed!" and nothing.
 */
void on_center_button() {
    static bool pressed = false;
    pressed = !pressed;
    if (pressed) {
        pros::lcd::set_text(2, "I was pressed!");
    } else {
        pros::lcd::clear_line(2);
    }
}

/**
 * Runs initialization code. This occurs as soon as the program is started.
 *
 * All other competition modes are blocked by initialize; it is recommended
 * to keep execution time for this mode under a few seconds.
 */
void initialize() {
    pros::lcd::initialize();
    pros::lcd::set_text(1, "Hello PROS User!");

    pros::lcd::register_btn1_cb(on_center_button);
}

/**
 * Runs while the robot is in the disabled state of Field Management System or
 * the VEX Competition Switch, following either autonomous or opcontrol. When
 * the robot is enabled, this task will exit.
 */
void disabled() {}

/**
 * Runs after initialize(), and before autonomous when connected to the Field
 * Management System or the VEX Competition Switch. This is intended for
 * competition-specific initialization routines, such as an autonomous selector
 * on the LCD.
```

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Welcome | main.cpp

src > main.cpp > opcontrol()

```cpp
#include "main.h"

void initialize() {
    pros::lcd::initialize();
    pros::lcd::set_text(1, "Hello PROS User!");

}

void autonomous() {}


void opcontrol() {


    while (true) {

    }
}
```

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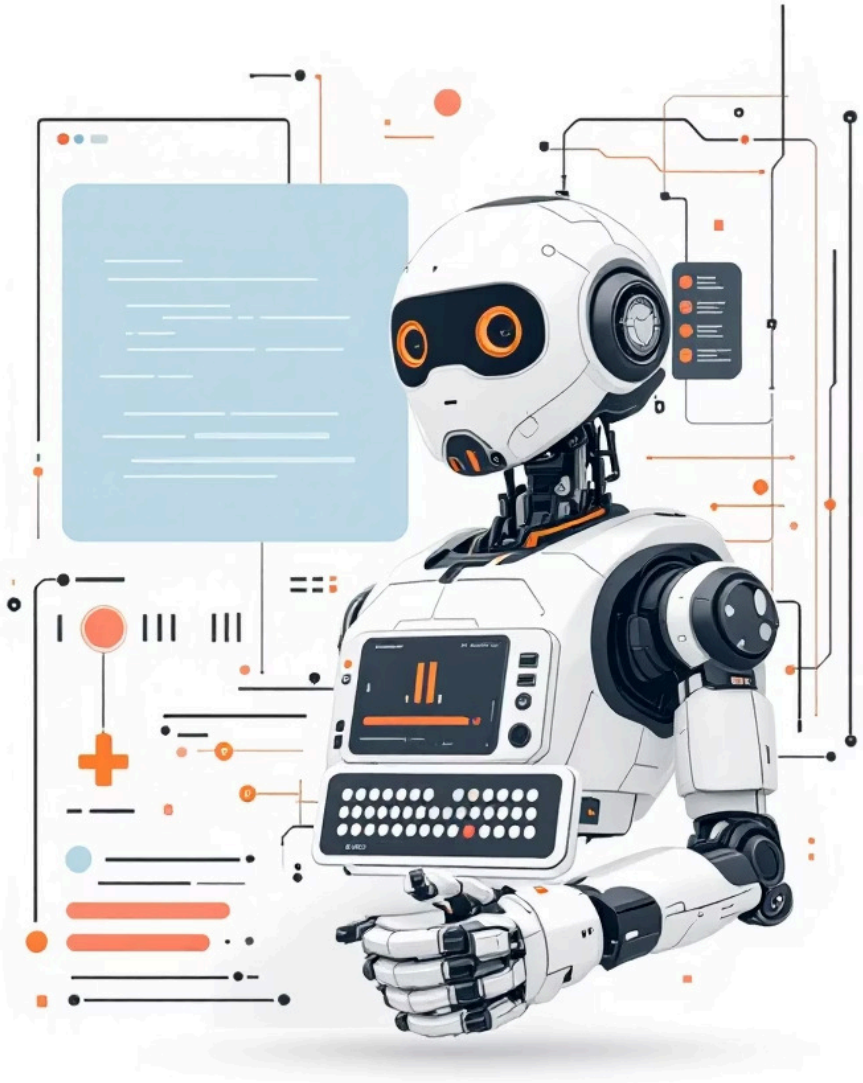

Hardware setup

DECLARAMOS MOTORES

```
pros::MotorGroup left_mg({8, 10});
pros::MotorGroup right_mg({1, 2});
```

DECLARAMOS IMU

```
pros::Imu imu(5);
```

DECLARAMOS EL CONTROL

```
pros::Controller master(pros::E_CONTROLLER_MASTER); // entrada control
```

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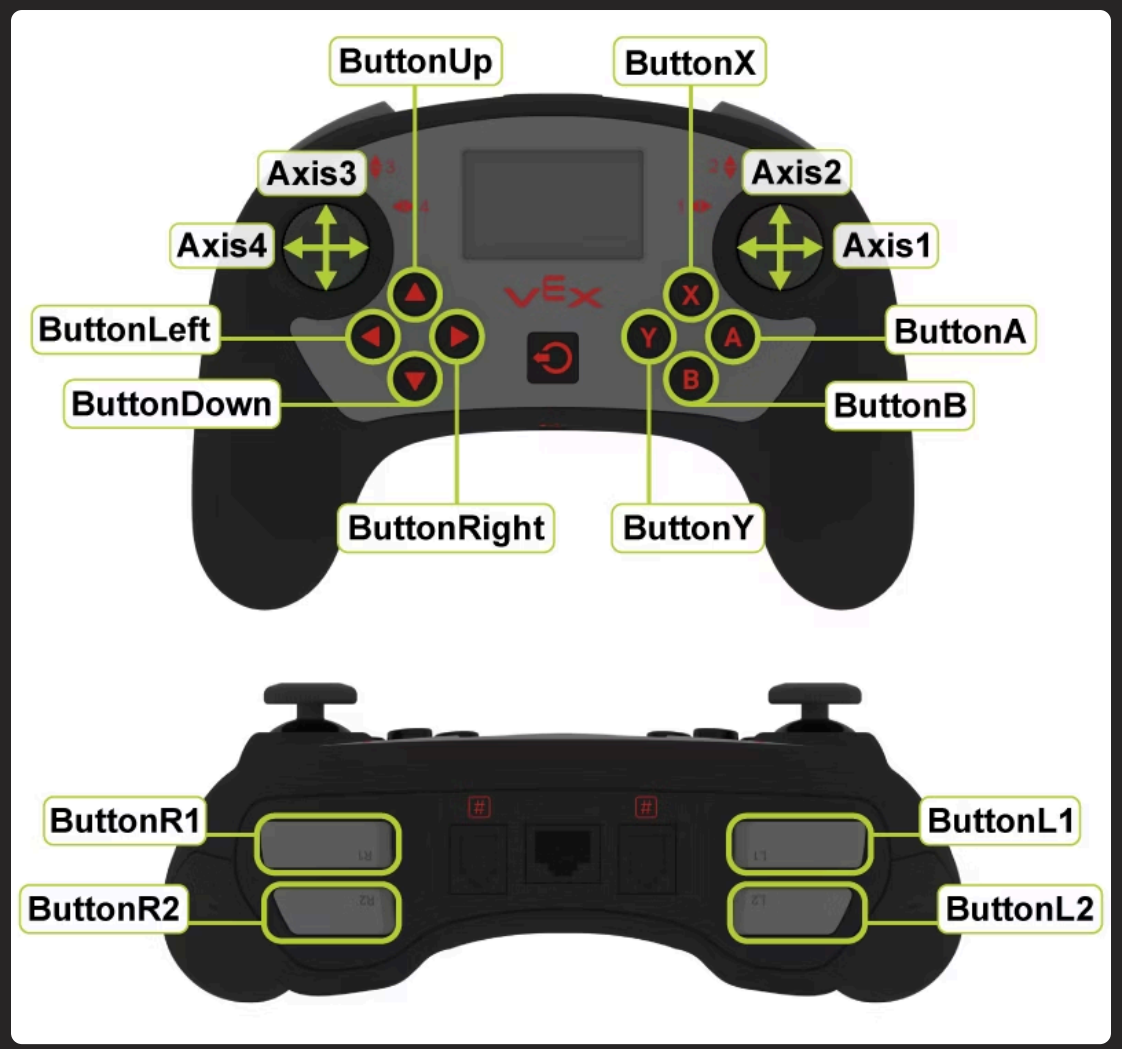
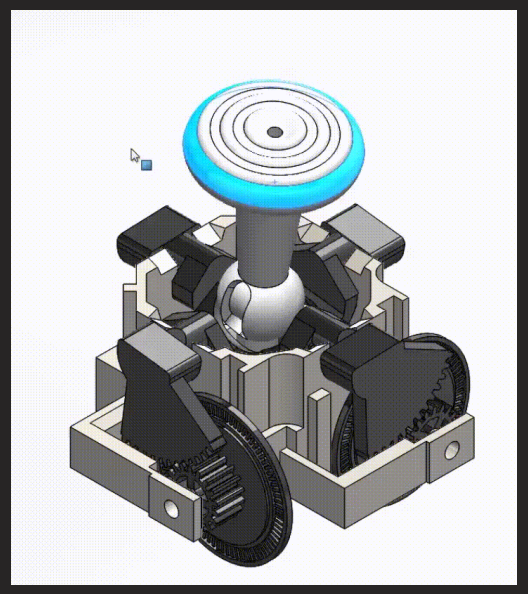


Joysticks (analógico)

```
master.get_analog(pros::E_CONTROLLER_ANALOG_RIGHT_Y); // avance/retroceso
master.get_analog(pros::E_CONTROLLER_ANALOG_LEFT_X); // giro
```



~ 127 → arriba / derecha

~ -127 → abajo / izq

0 → centro

- RIGHT_Y → avanzar/retroceder
- LEFT_X → girar

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Control manual (opcontrol)

```
void opcontrol() {}
```

Repetir el code

siempre hasta que salgas de opcontrol()

```
void opcontrol() {
    while (true) {
    }
}
```

power (±) = avanzar + / retroceder -

Definimos el stick derecho y eje vertical Y

```
int power  =  master.get_analog(pros::E_CONTROLLER_ANALOG_LEFT_Y);  // adelante/atrás
```

girar

Definimos el stick izq y el eje horizontal

```
int turn = -master.get_analog(pros::E_CONTROLLER_ANALOG_RIGHT_X); // girar
```

deadband

evita accidentes

```
if (std::abs(turn) < 5) turn = 0;
if (std::abs(power)  < 5) turn  = 0;
```

Si el valor es menor, tratarlo como si fuera 0.

mezcla

Para girar un lado del robot debe de empujar el otro.

```
int left_power  = power + turn;
int right_power = power - turn;
```

Avanzar Recto:

```
power = 80
turn = 0
...
left_pwr = 80 + 0 = 80
right_pwr = 80 - 0 = 80
```

Girar:

```
power = 0
turn = -80
...
left_pwr = 0 + 80 = 80
right_pwr = 0 - 80 = -80
```

Mover los motores:

se mueve de acuerdo a los joysticks:

```
left_mg.move(left_power);
right_mg.move(right_power);
```

```
void opcontrol() {
        while (true) {

                int power = -master.get_analog(pros::E_CONTROLLER_ANALOG_RIGHT_Y); // adelante/atrás
                int turn  =  master.get_analog(pros::E_CONTROLLER_ANALOG_LEFT_X);  // girar

                // Deadband
                if (std::abs(power) < 5) power = 0;
                if (std::abs(turn)  < 5) turn  = 0;

                // Mezcla
                int left_power  = power + turn;
                int right_power = power - turn;

                // Mover motores
                left_mg.move(left_power);
                right_mg.move(right_power);

                pros::delay(20);

                printIMU();
                // printMotors();
        }
}
```

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NOTIFICATIONS

Device VEX V5 Controller Port (COM5) has disconnected!

Project Uploaded Successfully

Made with GAMMA

project.pros

PROS Project Settings

Upload: **Project Name**

This shows as the program's name on the brain when uploaded.

test_robot

Upload: **Project Description**

This shows as the program's description on the brain when uploaded.

My PROS Project

Upload: **Program Slot**

Set the Program Slot number that this project will be uploaded into by default on the V5 brain.

1

Upload: **Program Icon**

Set the Program Icon do be displayed on the V5 Brain.

PROS

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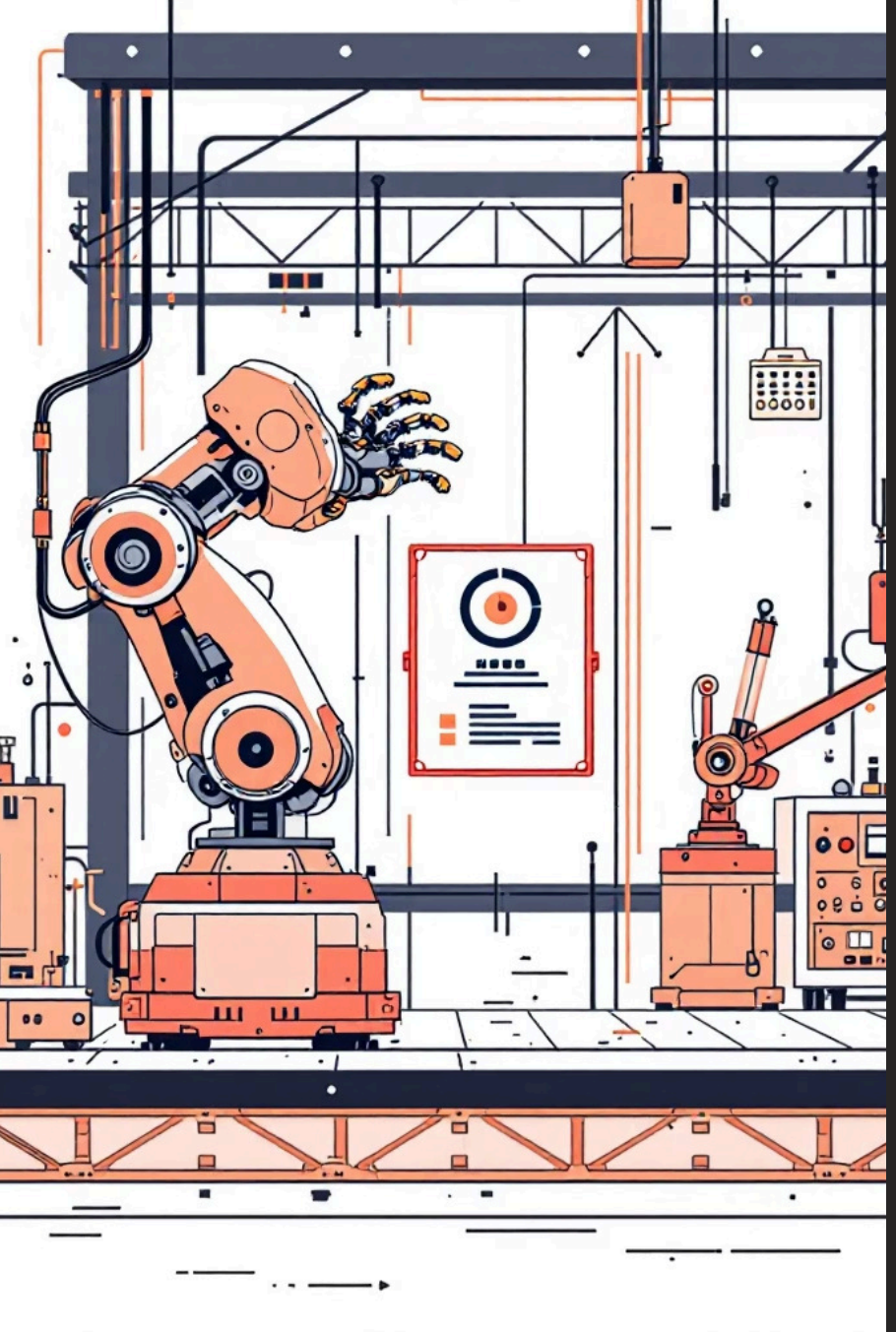

¿Por qué un robot falla aunque esté programado?

```
double error = target - imu.get_yaw(); // mide diferencia
double output = kp*error + ki*integral + kd*derivative; // calcula corrección
left.move(-output); // aplica movimiento
right.move(output);
```

El problema fundamental: Falta de control en tiempo real. El robot ejecuta comandos sin verificar si los está realizando correctamente.

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Sensores clave en VEX V5

IMU

Mide la orientación del robot.

Encoders

Miden la distancia recorrida.

Sin sensores = robot ciego.

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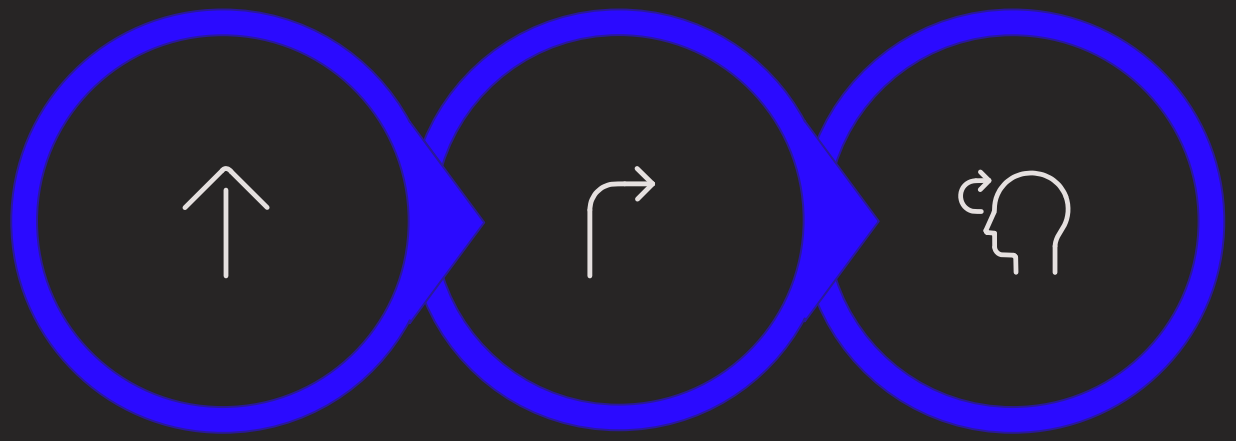
IMU: Medición de orientación



0° Adelante

90° Derecha

180° Atrás

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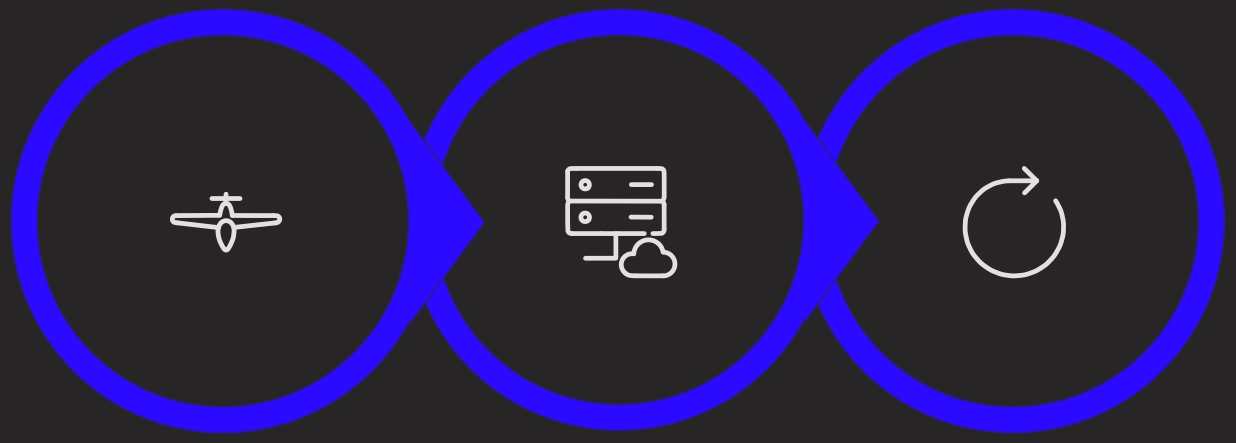
Encoders: Medición de movimiento



Motor girando

Obtener posición

360° = 1 rotación

Código

```
motor.get_position();
```

Concepto

Mide la rotación del eje en grados.

Ejemplo

360° = 1 rotación completa

Made with GAMMA

Print IMU

Declara el IMU

```
const int IMU_SIGN = 1;
```

Inicializamos y calibramos el IMU

```
void initialize() {
  pros::lcd::initialize();
  pros::lcd::set_text(1, "Hello PROS User!");

   imu.reset();
  while(imu.is_calibrating()){ pros::delay(10);}
}
```

```
    void printIMU() {
   static uint32_t last_print = 0;

   if (pros::millis() - last_print > 150) {
      last_print = pros::millis();

      int yaw = (int)(IMU_SIGN * imu.get_yaw());

      master.print(0, 0, "Yaw:%4d      ", yaw);
   }
}
```

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Print Motors

```
void printMotors() {
    static uint32_t last_print = 0;

    if (pros::millis() - last_print > 150) {
        last_print = pros::millis();

        int l_rpm = (int)left_mg.get_actual_velocity();
        int r_rpm = (int)right_mg.get_actual_velocity();

        master.print(1, 0, "L:%4d R:%4d  ", l_rpm, r_rpm);
    }
}
```

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Cómo llamarlas

```
printIMU();
printMotors();
```

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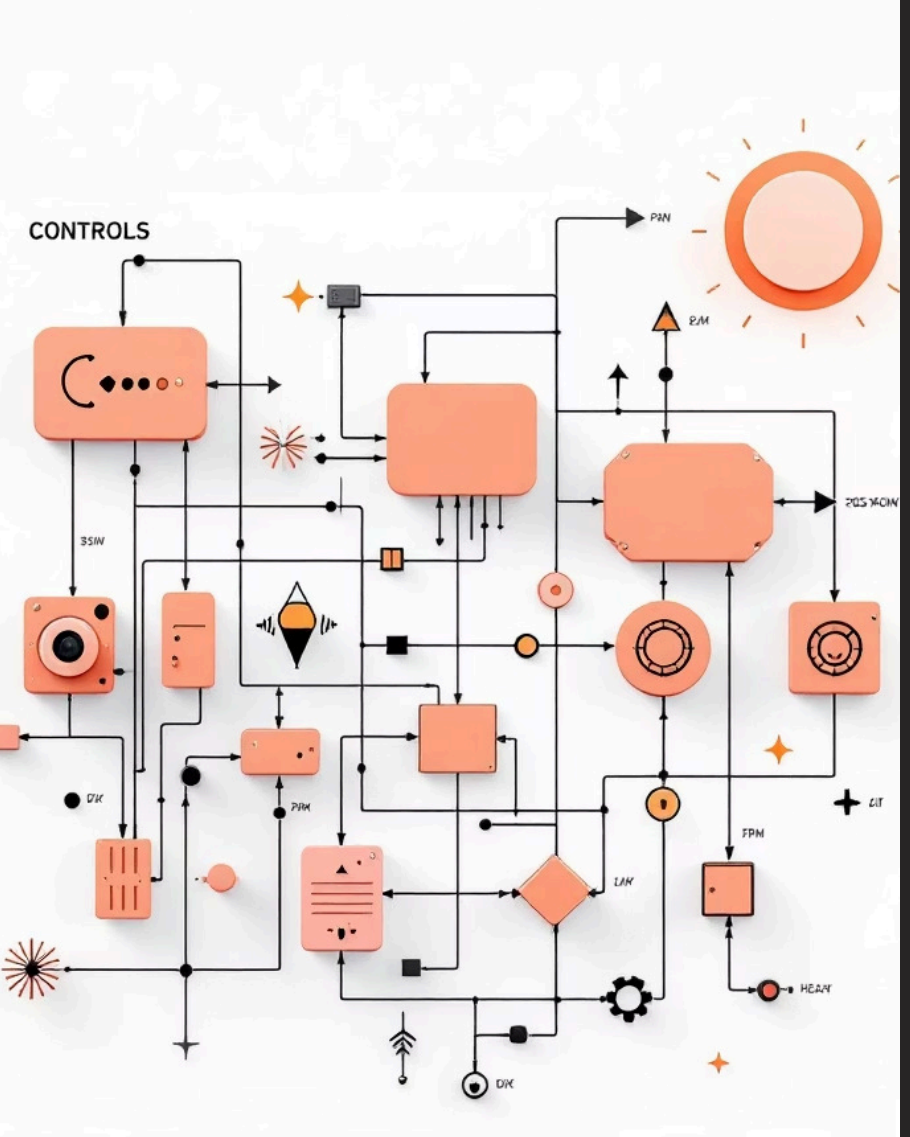

Un robot es un sistema de control

Sensor

Controlador

Actuador

Este ciclo se repite constantemente

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Conceptos fundamentales de control

Objetivo (Target)

Lo que queremos alcanzar.

Estado Actual (Actual)

Lo que está sucediendo ahora.

Error

La diferencia entre ambos.

`error = target - actual`

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Traducción simple

Sensor

¿Qué está pasando?

Error

¿Qué tan lejos estoy?

Control

¿Qué hago para corregir?

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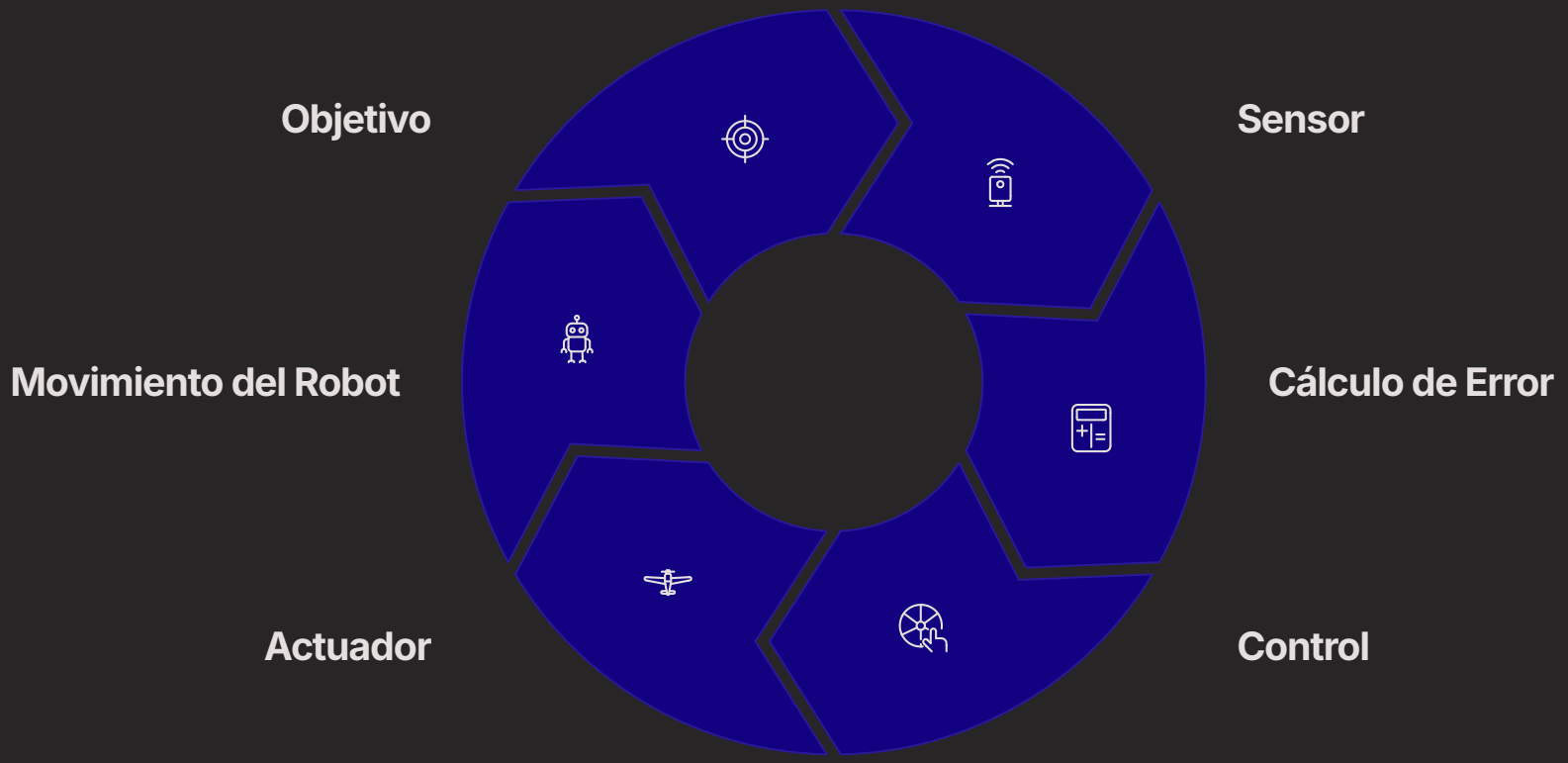
El ciclo completo del control



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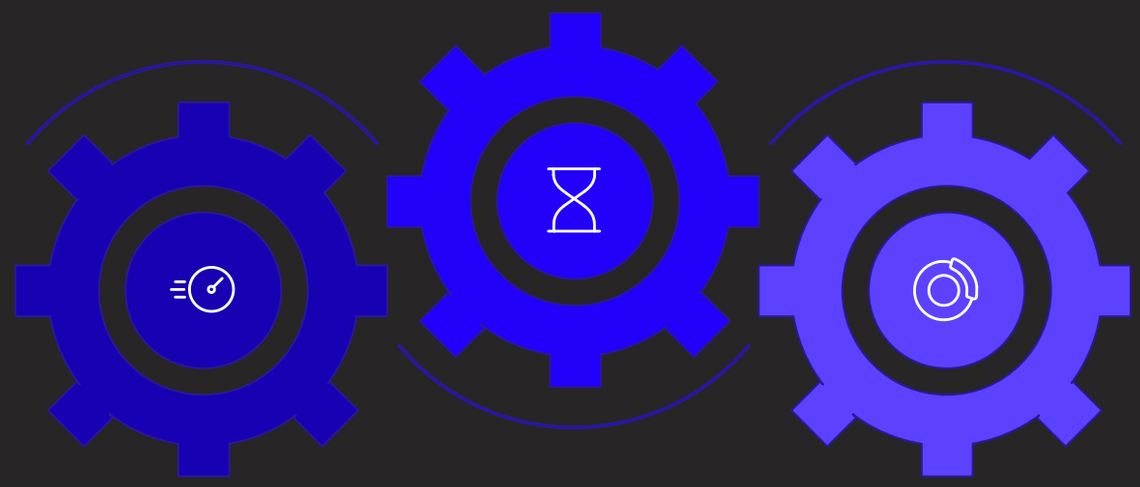
PID: Todo junto 🔥

I - Integrador

Empuje constante por errores acumulados



P - Proporcional

Corrección fuerte para errores grandes

D - Derivativo

Freno para evitar el sobrepaso

P te acerca, I te ajusta, y D te estabiliza.

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GIRAR

Resetear el IMU:

```
imu.tare_yaw(); // empezar en 0°
```

Velocidad fija:

```
const int speed = 50;
```

Leer el angulo:

```
double yaw = imu.get_yaw();
```

Condición para detenerse:

```
if (yaw >= angleDeg) break;
```

Movimiento de giro

```
left_mg.move(-speed);
right_mg.move(speed);
```

Detener motores

```
left_mg.brake();
right_mg.brake();
```

```
void turn(double angleDeg) {

    imu.tare_yaw(); // empezar en 0°

    const int speed = 50;

    while (true) {

        double yaw = imu.get_yaw();

        // cuando llega al ángulo → se detiene
        if (yaw >= angleDeg) break;

        // girar
        left_mg.move(-speed);
        right_mg.move(speed);

        pros::delay(20);
    }

    left_mg.brake();
    right_mg.brake();
}
```

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Drive

Definimos una vel

```
int speed = 60;
```

Movemos os motores a esa vel

```
left_mg.move(speed);
right_mg.move(speed);
```

Mientras se cumplan los seg definidos

```
uint32_t start_time = pros::millis();
while ((pros::millis() - start_time) < time_ms) {
```

```
// ================== DRIVE SIMPLE ==================
void drive(double time_ms) {

    int speed = 60;

    uint32_t start_time = pros::millis();

    while ((pros::millis() - start_time) < time_ms) {

        left_mg.move(speed);
        right_mg.move(speed);

        pros::delay(20);
    }

    left_mg.brake();
    right_mg.brake();
}
```

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Con PID

Variables con PID

```
double integral = 0;
double last_error = 0;
```

- integral » acumula error
- last_error » guardar error anterior

Contantes

```
const double kp = 2.7;
const double ki = 0.01;
const double kd = 1.4;
```

- kp » que tan fuerte corrige
- ki » corrige errores acumulados
- kd » evita que se pase

```
const int max_speed = 60;
const int min_speed = 15;
const int tolerance = 1;
```

- max_speed » velocidad max
- min_speed » vel minima
- tolerance » que tan cerca es suficiente

Control del tiempo

Guarda el tiempo de inicio para que se ejecute por max 4 s

```
uint32_t start_time = pros::millis();
```

```
while ((pros::millis() - start_time) < 4000) {}
```

Medicion y error

lee el error

```
double yaw = IMU_SIGN * imu.get_yaw();
double error = angleDeg - yaw;
```

calcula el error

Condición de paro

si ya estás cerca » se detiene

```
if (std::abs(error) <= tolerance) break;
```

- si ya estás cerca → se detiene

INTEGRAL

```
integral += error; if (integral > 100) integral = 100;
if (integral < -100) integral = -100;
```

• acumula el error con el tiempo • limita para que no crezca demasiado

DERIVADA

```
double derivative = error - last_error; last_error = error;
```

• mide qué tan rápido cambia el error

PID

```
double output = (kp * error) + (ki * integral) + (kd * derivative);
```

• combina todo el PID

• P → corrige • I → ajusta • D → frena

LÍMITES

```
if (output > max_speed) output = max_speed;
if (output < -max_speed) output = -max_speed;
```

• evita que el robot vaya demasiado rápido

```
if (std::abs(output) < min_speed) output = (output > 0) ? min_speed : -min_speed;
```

• asegura que el robot sí se mueva

MOVIMIENTO

```
left_mg.move(-output); right_mg.move(output);
```

• un lado avanza, el otro retrocede

DELAY

```
pros::delay(20);
```

• espera 20 ms

FINAL

```
left_mg.brake(); right_mg.brake();
```

• detiene el robot al terminal

```
// ================= TURN =================
void turn(double angleDeg) {
  imu.tare_yaw();

  double integral = 0;
  double last_error = 0;

  const double kp = 2.7;
  const double ki = 0.01;
  const double kd = 1.4;

  const int max_speed = 60;
  const int min_speed = 15;
  const int tolerance = 1;

  uint32_t start_time = pros::millis();

  while ((pros::millis() - start_time) < 4000) {

    double yaw = IMU_SIGN * imu.get_yaw();
    double error = angleDeg - yaw;

    if (std::abs(error) <= tolerance) break;

    integral += error;
    if (integral > 100) integral = 100;
    if (integral < -100) integral = -100;

    double derivative = error - last_error;
    last_error = error;

    double output = (kp * error) +
                    (ki * integral) +
                    (kd * derivative);

    if (output > max_speed) output = max_speed;
    if (output < -max_speed) output = -max_speed;

    if (std::abs(output) < min_speed)
      output = (output > 0) ? min_speed : -min_speed;

    left_mg.move(-output);
    right_mg.move(output);

    pros::delay(20);
  }

  left_mg.brake();
  right_mg.brake();
}
```

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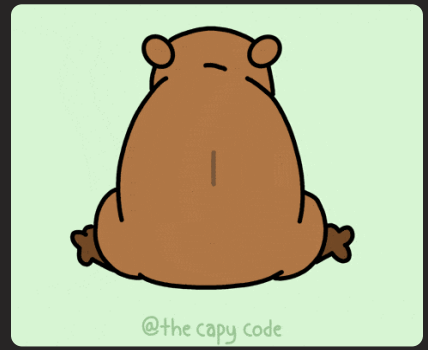
Se implementa




```
void autonomous() {
        turn(90);
        drive(2000); //avanza 2 seg
    }
```

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Tuning en la práctica 🎯

Ejemplo real:

Si mi robot gira a 90°

Solo P:

Se pasa a 100°, regresa a 80°, vibra

Con D:

Frena antes y llega más limpio

Con I mal ajustado:

Empieza a oscilar más

Por eso el tuning es importante. Cada robot es diferente.

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